

DEPENDENCE ON EARLY AND LATE REVERBERATION OF SINGLE-CHANNEL SPEAKER DISTANCE ESTIMATION

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ABSTRACT

Single-channel speaker distance estimation has recently achieved centimeter-level accuracy in simulated environments, yet it remains unclear which components of the room impulse response (RIR) the model exploits and how performance depends on the recording conditions. In this work, we decompose simulated RIRs into four variants (full, direct-only, no-late, and no-early) using the mixing time estimated from the echo density function as the boundary between early reflections and late reverberation. We define four calibration scenarios, from fully calibrated (synchronised capture, known source level) to fully uncalibrated (arbitrary onset, unknown level), and evaluate all combinations on a matched dataset. Results show that without time calibration, mean absolute error (MAE) increases to 1.29 m and the model extracts reverberation-based cues, with early reflections emerging as the most informative component. Further analysis against DRR, C_{50} , and T_{60} confirms that estimation accuracy improves with stronger early energy and degrades in highly reverberant environments. When time calibration is available, the model achieves a MAE of 0.14 m by extracting the propagation delay alone, regardless of the RIR content.

Index Terms— Acoustics, Speaker Distance Estimation, Deep Learning, Room Impulse Response

1. INTRODUCTION

Estimating the distance of a sound source from a single-channel recording is a problem of practical interest in hearing aid devices [1], hands-free communication [2], and speech recognition [3]. Unlike direction of arrival (DoA) estimation, which humans solve reliably through binaural cues such as interaural time difference (ITD) and interaural intensity difference (IID) [4], distance perception is inherently more difficult: binaural cues carry negligible distance information beyond close proximity [5], and listeners must instead rely on a combination of level attenuation, reverberation, and familiarity with the source signal [5, 6]. When these cues are degraded, human distance perception deteriorates considerably [7, 8].

Early approaches to source distance estimation operated on binaural recordings and relied on hand-crafted features derived from the direct-to-reverberant ratio (DRR) [9] or from inter-channel correlations. Vesa trained Gaussian mixture models (GMMs) on binaural correlation features to classify distance into discrete bins [10, 11], while in [12] the standard deviation of inter-channel differences is used with ensembles of GMMs and support vector machines (SVMs). These methods typically required careful hyperparameter tuning and generalised poorly across rooms with different acoustic conditions. More recent studies have revisited the problem with deep neural networks (DNNs), yet most remain

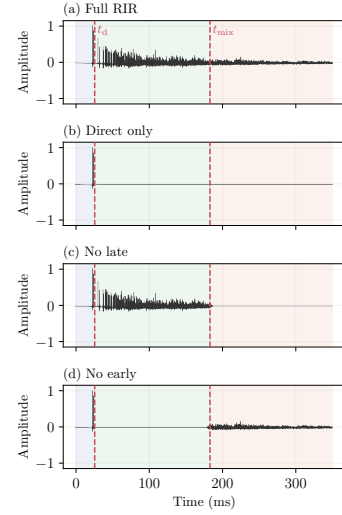


Fig. 1. RIR decomposition with source-microphone distance of $d = 7.2$ meters, $t_d = 25.4$ milliseconds and $t_{\text{mix}} = 182.9$ milliseconds.

limited to binary far/near classification [13, 14] or to coarse distance bins over a narrow range [8, 15, 16]. Single-channel setups, motivated by the constraints of low-power devices with limited computational resources [8, 17], have received comparatively little attention. The work in [18] was the first to frame speaker distance as a continuous regression problem, demonstrating that a DNN operating on short-time fourier transform (STFT) phase features can achieve centimeter-level accuracy in simulated environments. It is worth noting that the centimeter-level accuracy reported in [19, 18] was obtained on simulated data where the convolved signal preserves the propagation delay as a leading silence proportional to distance. In practice, however, the signal onset has no fixed relation to the source-microphone distance, making this cue a simulation artifact rather than a usable feature. In addition, that study treated the model as a black box and did not investigate which acoustic cues it leverages. Then, in [20] the authors proposed a data augmentation approach for generating realistic room-acoustic conditions to support speaker distance estimation using the convolutional recurrent neural network (CRNN) model in [19].

These findings raise a natural question that remains unanswered: *which temporal components of the RIR does the model actually exploit, and to what extent does each component contribute to estimation accuracy?* A RIR between a source and a microphone can be decomposed into three successive regions: the direct path, early reflections, and late reverberation, each carrying qualitatively different

information about the source-microphone geometry and the enclosing room. The direct path encodes distance through propagation delay and $1/r$ attenuation, with r being the source-receiver distance; early reflections add geometric constraints imposed by nearby wall surfaces; late reverberation provides a statistical fingerprint of the room’s volume and absorption. Understanding which of these regions drives a learned estimator is essential both for interpreting the model’s behaviour and for guiding the design of more robust systems.

In this work, we present a systematic analysis of the single-channel speaker distance estimation task under controlled calibration conditions. Our contributions are as follows:

- We decompose simulated RIRs into four variants (full, direct-only, no-late, and no-early) using the mean-free-path mixing time [21, 22] as the boundary between early reflections and late reverberation, and train separate models on each variant to isolate the contribution of every temporal region.
- We modeled two distance cues beyond reverberation. The first is the propagation delay preserved in simulated data. The second is the source emission level, which in some practical deployments (e.g., a loudspeaker at fixed output) provides a usable amplitude cue via the $1/r$ law. Crossing onset randomisation with level variation on a matched dataset yields four conditions that isolate the contribution of each cue.
- We present a 4×4 evaluation (four calibration conditions with four RIR variants) that reveals how much of the estimation accuracy is attributable to calibration cues versus reverberation-based acoustic features, and which RIR components compensate when calibration information is unavailable.

The work is structured as follows: Section 2 provides all the details regarding the generated dataset (including all RIR variant and calibration scenarios) and the employed baseline model; Section 3 depicts all the results in all conditions whereas Section 4 draws the conclusions.

2. EXPERIMENTAL SETUP

We consider a single omnidirectional microphone placed at an unknown position $\mathbf{m} \in \mathbb{R}^3$ inside a room, capturing the speech signal of a talker located at an unknown position $\mathbf{s} \in \mathbb{R}^3$ in the same unknown room. Neither the room geometry, the acoustic properties, nor the absolute positions of the source and microphone are assumed to be known. The goal is to estimate the distance $r = \|\mathbf{s} - \mathbf{m}\|$ from the recorded single-channel signal alone.

2.1. Dataset

Anechoic speech recordings obtained from the EARS dataset [23] are convolved with the simulated RIRs from *pyroomacoustics* [24]. Table 1 depicts the range of random parameters for data generation, following [18, 19]. The experiments include 2500 audio files of 10 s duration at 16 kHz. The samples are randomly assigned to 5 folds to assess the performance in a 5-fold cross-validation fashion. By doing so, each cross-fold iteration assigns 1500, 500, and 500 audios to training, validation, and testing sets, respectively. In this study, we impose the distribution of the distances to be uniform between 1 and 11 meters to avoid biasing the distance estimator with respect to short distances. This is done by generating room and source-receiver configurations that match the target distance. Detailed statistics of the synthetic dataset are shown in Figure 2.

Table 1. Parameters for data generation.

Parameter	Random ranges
Room width and length	[3.0, 15.0] m
Room height	[2.0, 7.0] m
# of materials (wall, floor, ceiling)	13, 7, 8
Source - receiver height	[1.5, 2.2] m
Source-to-surface distance	> 0.5 m
Source-to-receiver distance	> 1.0 m

Table 2. Calibration scenarios. ✓: calibration cue available; ✕: unavailable (simulated via randomisation).

Scenario	Time-cal.	Level-cal.	Condition
FULLY CALIBRATED	✓	✓	Synchronised capture, known source level
TIME-CALIBRATED	✓	✕	Synchronised capture, unknown level
LEVEL-CALIBRATED	✕	✓	Arbitrary onset, known source level
UNCALIBRATED	✕	✕	Arbitrary onset, unknown level

2.2. Room impulse response decomposition

A RIR between a source at position $\mathbf{s}$ and a microphone at position $\mathbf{m}$ can be decomposed into three temporally non-overlapping regions:

$$h(t) = \underbrace{h_d(t)}_{\text{direct path}} + \underbrace{h_e(t)}_{\text{early reflections}} + \underbrace{h_l(t)}_{\text{late reverberation}}, \quad (1)$$

where each component occupies a successive time interval. The direct-path component h_d is a delayed and attenuated impulse whose time of arrival $\tau_d = \|\mathbf{s} - \mathbf{m}\|/c$ is determined by the source-microphone distance and the speed of sound $c \approx 343 \text{ m s}^{-1}$. Early reflections $h_e(t)$ consist of specular, sparse wall bounces that carry geometric information about the room. Late reverberation $h_l(t)$ arises once reflections overlap sufficiently to form a diffuse, stochastic field.

The mixing time t_{mix} marks the onset of diffuse reverberation, i.e., the point beyond which the RIR is well modeled by a Gaussian noise process [22]. For each RIR, we compute the normalised echo density $\eta(t)$ of Abel and Huang [22], and selecting $t = t_{\text{mix}}$ such that $\eta(t) \approx 1$. Using the direct-path endpoint $t_d = \tau_d + \Delta t_d$ (with $\Delta t_d = 2 \text{ ms}$ guard window) and the mixing time t_{mix} as boundaries, we derive four RIR variants from each simulated impulse response by applying smooth half-cosine fade-outs and fade-ins (5 ms duration) at the relevant boundaries:

1. **Full.** Raw RIR $h(t)$;
2. **Direct only.** $h(t)$ faded to zero after t_d , retaining only the direct path;
3. **No late.** $h(t)$ faded to zero after t_{mix} , removing the late reverberant tail;
4. **No early.** The direct path up to t_d plus the late reverberation from t_{mix} onward with early reflections suppressed.

These variants allow us to isolate the contribution of each temporal region to distance estimation. They can be visually inspected in Figure 1. By training and evaluating on every variant, we can determine whether the model relies primarily on direct-path attenuation, early-reflection patterns, or late reverberation statistics.

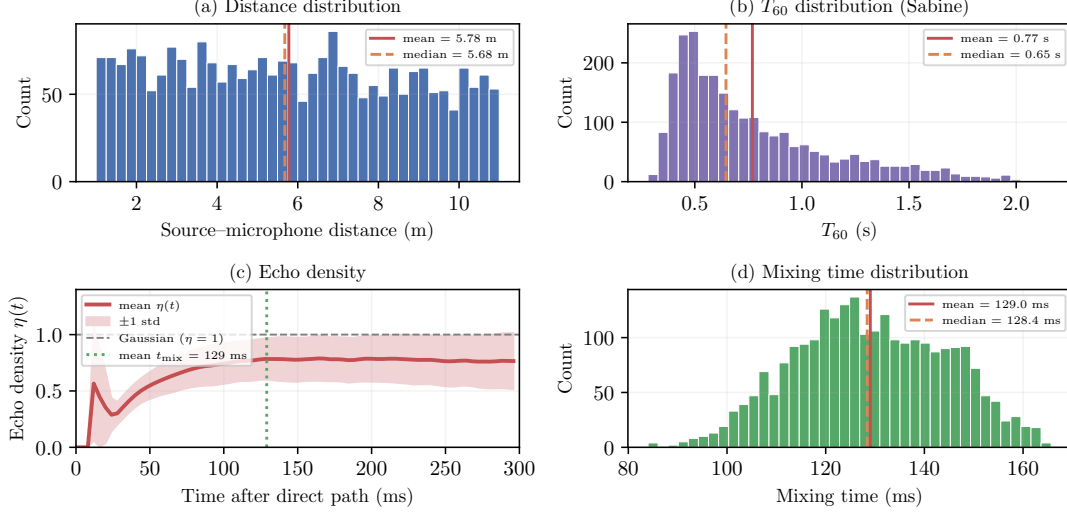


Fig. 2. Dataset statistics (shared across all four scenarios, $N = 2500$ samples). (a) Source-microphone distance distribution ($r \in [1, 14]$ m, uniform sampling). (b) Reverberation time T_{60} estimated via Sabine’s formula. (c) Normalised echo density $\eta(t)$ [22]; the vertical line marks the mean mixing time t_{mix} , confirming that $\eta \approx 1$ (Gaussian statistics) at the chosen early/late boundary. (d) Distribution of the mixing time t_{mix} .

2.3. Calibration scenarios

The cues available for distance estimation depend on the degree of calibration of the recording setup. We identify two calibration dimensions: **Time calibration.** Simulated RIRs introduce a leading silence because of synchronised capture between microphone and source, whose duration is equal to the propagation delay $\tau_d = r/c$, encoding distance directly. This cue is an artifact of the data simulation pipeline as in no real recordings the signal and the recording are perfectly time-calibrated. To remove it, we strip the natural delay from the convolved output and prepend a random silence $\delta \sim \mathcal{U}\{0, \lfloor r_{\text{max}}/(c/f_s) \rfloor\}$. The RIR itself is never modified. **Level calibration.** When the source emission level is known, the received energy decreases with distance via the $1/r^2$ law, providing a direct amplitude cue. In an uncalibrated scenario, different speakers produce varying levels, which breaks this correlation. We simulate this by applying a random gain $G \sim \mathcal{U}(-6, 6)$ dB to the anechoic speech before convolution.

Crossing the two calibration dimensions yields four scenarios (Table 2). All four share the *identical* set of rooms, source-microphone configurations, RIRs, talkers, and speech segments. By doing so, any performance difference is attributable solely to the available calibration information.

2.4. Baseline model

To estimate the speaker distance from single-channel recordings, we employ the state-of-the-art CRNN used in [19, 18]. Within each dataset condition, the model is trained and evaluated on all four RIR variants (full, direct-only, no-late, no-early), yielding a 4×4 results matrix that disentangles the effect of calibration scenarios from the contribution of each temporal region of the RIR. The performance evaluation of the distance estimator is carried out by computing the mean absolute error (MAE) $\mathcal{L}_1(y, \hat{y}) = |y - \hat{y}|$, where $y \in \mathbb{R}^+$ and $\hat{y} \in \mathbb{R}^+$ are the ground truth and predicted distances, respectively. In addition, following [19], we also compute the relative

$$\text{MAE } r\mathcal{L}_1(y, \hat{y}) = \frac{\mathcal{L}_1(y, \hat{y})}{y} = \frac{|y - \hat{y}|}{y}.$$

3. EXPERIMENTAL RESULTS

In this section a study on which temporal component of the RIR is used in each scenario is analyzed. Further analysis is carried out on the effect of propagation delay and source level for the speaker distance estimation task.

3.1. Dependence on RIR components

Figure 3 presents the performance of all four RIR variants under the uncalibrated scenario, which reflects the most common practical condition. The no-early variant, which retains the direct path and late tail but suppresses early reflections, performs worst, even below direct-only. This identifies early reflections as the single most informative component for distance estimation when calibration cues are unavailable. Moreover, the no-late variant nearly performs as the full version, indicating that the combination of direct sounds and early reflections are the most important ones for distance estimation.

The ribbon plots (b.1–b.4) confirm this across acoustic conditions. Error decreases with increasing DRR and C_{50} for the full and no-late variants, but the direct-only and no-early variants remain at a higher error level regardless, indicating they lack the distance information. At very low DRR (< -10 dB), all variants converge to similarly high error. Increasing T_{60} mildly degrades all variants, suggesting that while reverberation provides distance cues, excessive reverberation smears the temporal structure that carries them.

3.2. Effect of propagation delay and source level

Table 3 also reveals why prior works [19, 18] on simulated data achieved centimeter-level accuracy: the time-calibrated rows show that the propagation delay alone yields $\mathcal{L}_1 \approx 0.14$ m regardless of RIR content. Removing level calibration has negligible effect, confirming that the amplitude cue is redundant when timing is avail-

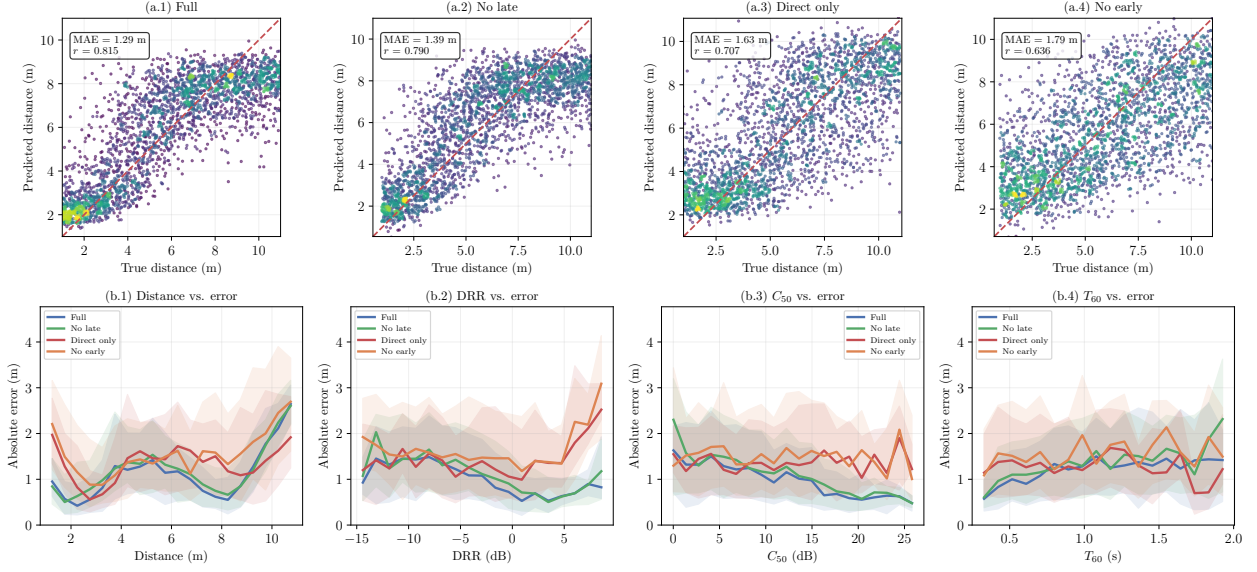


Fig. 3. Performance analysis under the uncalibrated scenario across all four RIR variants. **Top row** (a.1–a.4): predicted vs. true distance for each variant, with MAE and Pearson correlation r reported. Point colour encodes local density. **Bottom row** (b.1–b.4): median absolute error (solid lines) with interquartile range (shaded bands) as a function of source-microphone distance, DRR, C_{50} , and T_{60} .

Table 3. MAE with 95% confidence intervals and relative MAE (%) for each combination of calibration scenario and RIR variant. The best result per row is shown in **bold**. The *Random* row reports the expected error of a model that has been trained with silent recordings exploiting only distance distribution, serving as a sanity check.

Scenario	Time	Level	RIR variant			
			Full	Direct	No late	No early
Fully calibrated	✓	✓	0.15 [0.14, 0.16] (3.5%)	0.14 [0.13, 0.16] (3.4%)	0.15 [0.14, 0.16] (3.4%)	0.15 [0.13, 0.16] (3.5%)
Time-calibrated	✓	×	0.15 [0.13, 0.17] (3.5%)	0.14 [0.12, 0.15] (3.2%)	0.16 [0.14, 0.17] (3.7%)	0.15 [0.14, 0.16] (3.5%)
Level-calibrated	×	✓	1.29 [1.17, 1.42] (29.0%)	1.58 [1.52, 1.64] (38.3%)	1.38 [1.28, 1.48] (31.0%)	1.79 [1.65, 1.94] (45.3%)
Uncalibrated	×	×	1.29 [1.17, 1.41] (29.4%)	1.63 [1.52, 1.73] (41.3%)	1.39 [1.28, 1.50] (31.5%)	1.79 [1.65, 1.94] (44.9%)
Random	–	–	2.49 [2.43, 2.55] (72.2%)			

able. Without the propagation delay, $\mathcal{L}_1$ increases by an order of magnitude. Level calibration again provides almost no benefit, indicating that the $1/r$ amplitude cue is a weak predictor compared to reverberation-based features.

All conditions remain well above the random baseline ($\mathcal{L}_1 = 2.49$ m), confirming that the model extracts meaningful acoustic information even in the fully uncalibrated scenario.

4. CONCLUSIONS

This work presented a systematic analysis of how individual RIR components contribute to single-channel speaker distance estimation under different calibration conditions. When time calibration is available, the propagation delay alone yields $\text{MAE} \approx 0.14$ m and

all RIR components become redundant. Level calibration provides negligible benefit in every condition. In the uncalibrated scenario, the model shifts to reverberation-based cues, with early reflections emerging as the most informative component: removing them degrades performance to $\text{MAE} = 1.79$ m, worse than using the direct path alone (1.63 m), while retaining them without the late tail (1.39 m) nearly matches the full RIR (1.29 m). Further correlation analysis against DRR, C_{50} , and T_{60} confirms that estimation quality is based on the balance between early and reverberant energy, with excessive reverberation degrading all variants. Future work will investigate the robustness of these findings under additive noise conditions, extend the framework to moving speakers with time-varying source-microphone distances, and explore multi-microphone configurations where spatial cues may complement the reverberation-based features identified here.

5. REFERENCES

- [1] V. Hamacher, J. Chalupper, J. Eggers, E. Fischer, U. Kornagel, H. Puder, and U. Rass, "Signal processing in high-end hearing aids: State of the art, challenges, and future trends," *EURASIP Journal on Advances in Signal Processing*, vol. 2005, no. 18, pp. 1–15, 2005.
- [2] S. Oh, V. Viswanathan, and P. Papamichalis, "Hands-free voice communication in an automobile with a microphone array," in *IEEE International Conference on Acoustics, Speech, and Signal Processing*, 1992.
- [3] M. Omologo, P. Svaizer, and M. Matassoni, "Environmental conditions and acoustic transduction in hands-free speech recognition," *Speech Communication*, vol. 25, no. 1-3, pp. 75–95, 1998.
- [4] M. Dietz, S. D. Ewert, and V. Hohmann, "Auditory model based direction estimation of concurrent speakers from binaural signals," *Speech Communication*, vol. 53, no. 5, pp. 592–605, 2011.
- [5] P. Zahorik, D. S. Brungart, and A. W. Bronkhorst, "Auditory distance perception in humans: A summary of past and present research," *ACTA Acustica united with Acustica*, vol. 91, no. 3, pp. 409–420, 2005.
- [6] C. Mendonça, P. Mandelli, and V. Pulkki, "Modeling the perception of audiovisual distance: Bayesian causal inference and other models," *PloS one*, vol. 11, no. 12, p. e0165391, 2016.
- [7] E. C. Cherry, "Some experiments on the recognition of speech, with one and with two ears," *The Journal of the Acoustical Society of America*, vol. 25, no. 5, pp. 975–979, 1953.
- [8] E. Georganti, T. May, S. van de Par, A. Harma, and J. Mourjopoulos, "Speaker Distance Detection Using a Single Microphone," *IEEE Transactions on Audio, Speech, and Language Processing*, vol. 19, no. 7, pp. 1949–1961, 2011.
- [9] Y. Lu and M. Cooke, "Binaural Estimation of Sound Source Distance via the Direct-to-Reverberant Energy Ratio for Static and Moving Sources," *IEEE Transactions on Audio, Speech, and Language Processing*, vol. 18, no. 7, pp. 1793–1805, 2010.
- [10] S. Vesa, "Sound Source Distance Learning Based on Binaural Signals," in *IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*, 2007.
- [11] —, "Binaural Sound Source Distance Learning in Rooms," *IEEE Transactions on Audio, Speech, and Language Processing*, vol. 17, no. 8, pp. 1498–1507, 2009.
- [12] E. Georganti, T. May, S. van de Par, and J. Mourjopoulos, "Sound Source Distance Estimation in Rooms based on Statistical Properties of Binaural Signals," *IEEE Transactions on Audio, Speech, and Language Processing*, vol. 21, no. 8, pp. 1727–1741, 2013.
- [13] K. Patterson, K. Wilson, S. Wisdom, and J. R. Hershey, "Distance-Based Sound Separation," in *Interspeech*, 2022.
- [14] D. A. Krause, A. Politis, and A. Mesaros, "Joint direction and proximity classification of overlapping sound events from binaural audio," in *IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*, 2021.
- [15] M. Yiwere and E. J. Rhee, "Sound source distance estimation using deep learning: An image classification approach," *Sensors*, vol. 20, no. 1, p. 172, 2019.
- [16] A. Sobhdel, R. Razavi-Far, and V. Palade, "A Few-Shot Learning Approach for Sound Source Distance Estimation Using Relation Networks," in *International Conference on Machine Learning and Applications (ICMLA)*, 2024.
- [17] R. Venkatesan and A. Ganesh, "Analysis of monaural and binaural statistical properties for the estimation of distance of a target speaker," *Circuits, Systems, and Signal Processing*, vol. 39, p. 3626–3651, 2020.
- [18] M. Neri, A. Politis, D. A. Krause, M. Carli, and T. Virtanen, "Single-Channel Speaker Distance Estimation in Reverberant Environments," in *IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA)*, 2023, pp. 1–5.
- [19] —, "Speaker Distance Estimation in Enclosures From Single-Channel Audio," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 32, pp. 2242–2254, 2024.
- [20] J. Lin, G. Götz, H. S. Llopis, H. Hafsteinsson, S. Guðjónsson, D. G. Nielsen, F. Pind, P. Smaragdis, D. Manocha, J. Hershey, T. Kristjansson, and M. Kim, "Generative data augmentation challenge: Synthesis of room acoustics for speaker distance estimation," in *IEEE International Conference on Acoustics, Speech and Signal Processing Workshops (ICASSPW)*, 2025.
- [21] H. Kuttruff, *Room acoustics*. Crc Press, 2016.
- [22] J. S. Abel and P. Huang, "A simple, robust measure of reverberation echo density," in *Audio Engineering Society Convention 121*. Audio Engineering Society, 2006.
- [23] J. Richter, Y. Wu, S. Krenn, S. Welker, B. Lay, S. Watanabe, A. Richard, and T. Gerkmann, "EARS: An Anechoic Fullband Speech Dataset Benchmarked for Speech Enhancement and Dereverberation," in *Interspeech*, 2024.
- [24] R. Scheibler, E. Bezzam, and I. Dokmanić, "Pyroomacoustics: A python package for audio room simulation and array processing algorithms," in *IEEE ICASSP*, 2018.